24 A As the top pan balance's reading increases when there is a current in direction YX in the wire, there must be a force pushing the horseshoe magnet down against the balance. By Newton's 3rd Law, the magnetic force on the wire must be upwards. (There is a repulsion force between the wire and magnet)

By applying FLHR on the wire, the direction of magnetic flux density due to the horseshoe magnet can be determined to be coming out of face A. Thus, face A of the horseshoe magnet must be a North pole.

25 C Direction of magnetic flux density due to current in straight wire is determined by using RHGR